SIMATSENGINEERING(SAVEETHASCHOOLOFENGINEERING)

DepartmentofComputerScienceandEngineering

ASSESSMENTTOOL1–INTERACTIVETREE SIMULATOR

CourseCode:	CSA0305	**CourseTitle:**	DataStructures
COAssessed:	CO3–Precision(BL3,BL5)	**Assessment:**	AssessmentTool1–Interactive Tree Simulator
Weightage:	30%ofCIA	**TotalMarks:**	100
CO3Statement:	*SolveproblemsforoptimizeddatastorageandretrievalusingBinarySearchTrees,AVLTrees,B-Trees, Red-Black Trees, and Splay Trees.*		
Student Name:	VAISHNAV.M	**Reg. No.:**	192524251
Section/Batch:	2025	**Date:**	29/07/2026

CodeofConduct

I _____VAISHNAV M.(192524251)____ Name/RegNo) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature ofthe Student: VAISHNAV.M

Instructions

- Thisassessmenttoolcontains5InteractiveTreeSimulatorquestions,Total:100Marks.
- Studentscanuseanyonline/offlineInteractiveTreeSimulator.Demonstrateinsertion,deletion, searching, and traversal operations wherever applicable.
- Whenastudentanswerusingsimulator,inevaluationtheanswershouldhavecapturedscreenshots after every major operation.
- SubmitthereportinPDFformatwithproperlylabelledscreenshotsasproofof completion.

Section/Topic	Focus	QNos	Marks
BinarySearchTree	Properties,binarytreeand binarysearchtree, insertionand deletion	1	20
Tree Traversals	Traversaltypes,Inorder,PostorderandPreorder	2	20
AVL Tree	Balancingfactor,Rotation,Typesofrotation,examples	3	20
BTree,2-3tree,2-3-4tree	Properties,structuresandOperations	4	20
RedBlackTree	Properties,ExampleandOperations	5	20
		GRANDTOTAL:	**100Marks**

AnnexureA–InteractiveTreeSimulator

1. ConstructaBinarySearchTree(BST)byinsertingthefollowingkeys: 50,30,70,20,40,60,80

   Perform:

   a. InorderTraversal
   b. Searchfor 60
   c. Delete30
   d. DisplaythefinalBST.
2. CompleteTreeTraversalsbyInserting50,30,70,20,40,60,80.Findallthetraversals.
   a. Inorder
   b. Preorder
   c. Postorder
3. Insert70,50,90,40,60,80,100,55intoanAVLTree.Delete40and90.Analyzehowthetree rebalances after each deletion.
4. ConstructaB-Treeoforder3byinserting45,25,65,15,35,55,75
5. Insert100,80,120,60,90,110,130intoaRed-BlackTree.Analyzetheirheights.

RubricsforCalculation

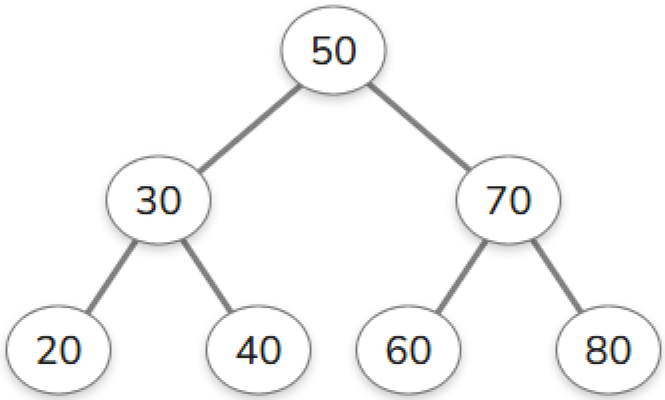
Criteria	Weightage	Level4–Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
CorrectTreeConstruction	20	Tree is constructedcorrectly with alloperationsperformedaccurately	Minorerrorsinconstruction	Someoperationsincorrect	Treeconstruction mostlyincorrect
UnderstandingofTreeOperations	20	Clearlyexplainsinsertion,deletion, search andbalancingoperations	Explainsmost operationscorrectly	Limitedunderstanding	Unabletoexplainoperations
AnalysisofTree Behaviour	30	Correctlyanalysesbalancing, rotations,nodesplit/merge,splaying	Goodanalysis withminoromissions	Partialanalysis	Poororincorrectanalysis
Presentation	20	Well-organizedreportwithaccurate observations andconclusion	Goodreportwithminorissues	Basic report withlimitedobservations	Incompletereport
Accuracyof Responses	10	90–100%correctanswers	75–89%correctanswers	50–74%correctanswers	Below50%correctanswers

FacultyIn-charge	**CourseCoordinator**	**ProgramDirector**
Signature: ____________	Signature: ____________	Signature: ____________
Date: ________________	Date: ________________	Date: ________________

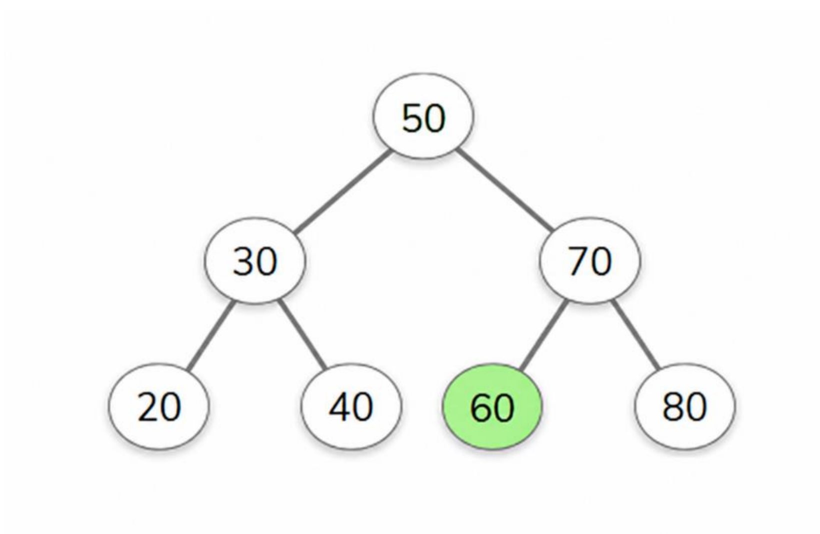
1. ConstructaBinarySearchTree(BST)byinsertingthefollowingkeys: 50,30,70,20,40,60,80

   Perform:

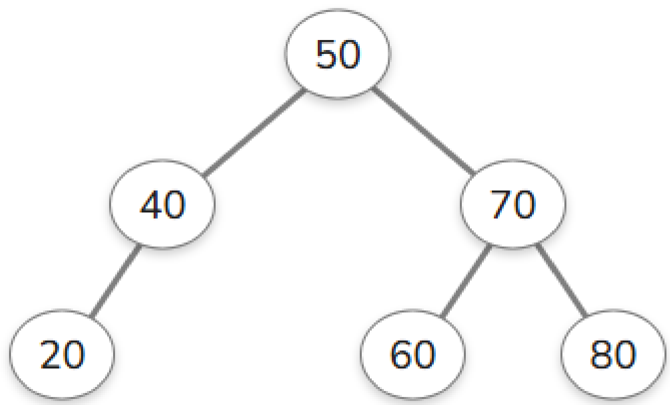
   a. InorderTraversal



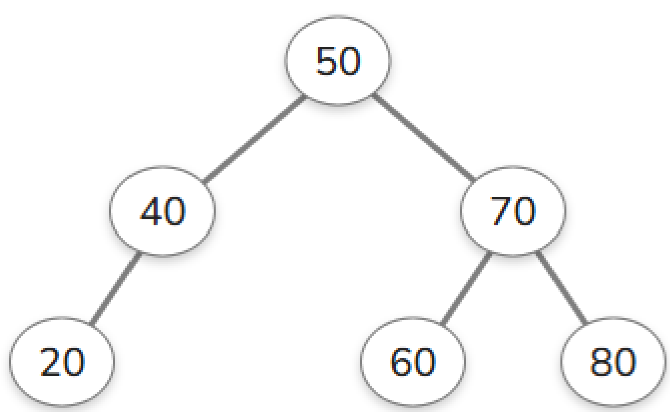
   b. Searchfor 60



c. Delete30

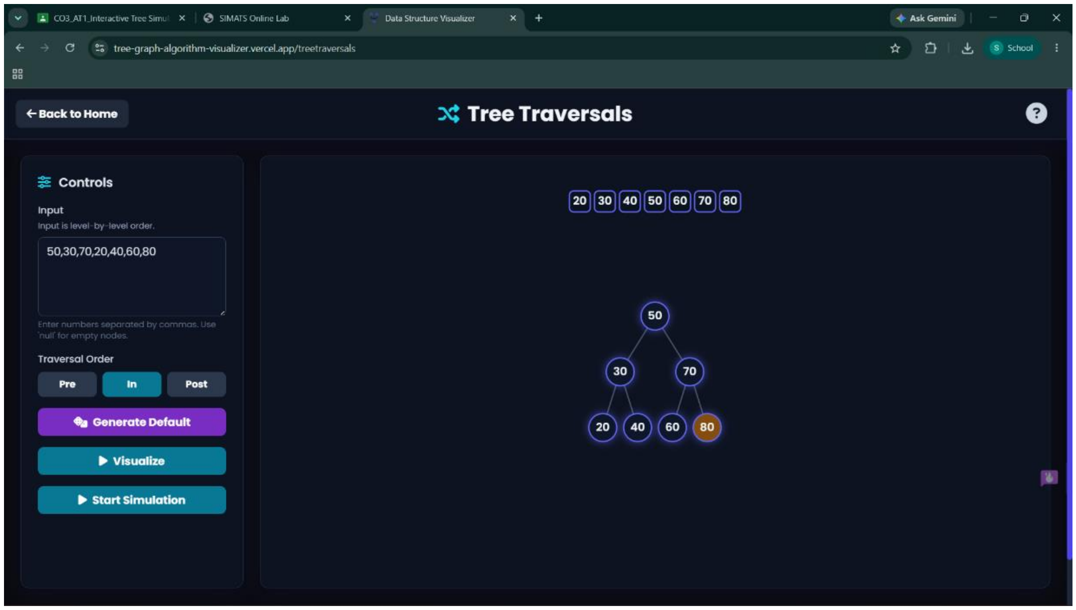


d.DisplaythefinalBST.

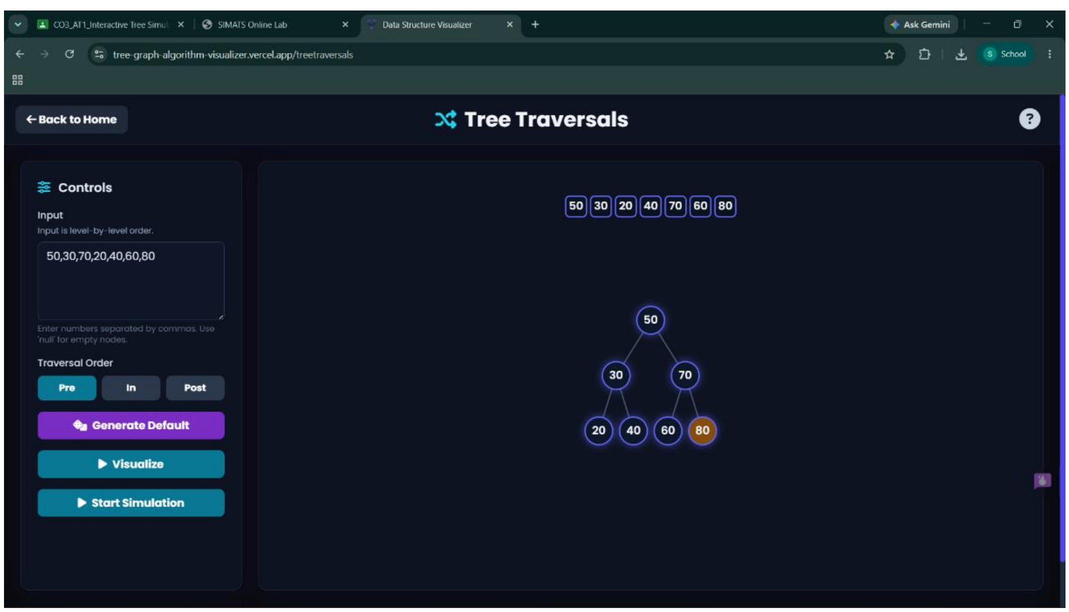


2.CompleteTreeTraversalsbyInserting50,30,70,20,40,60,80.Findallthetraversals.

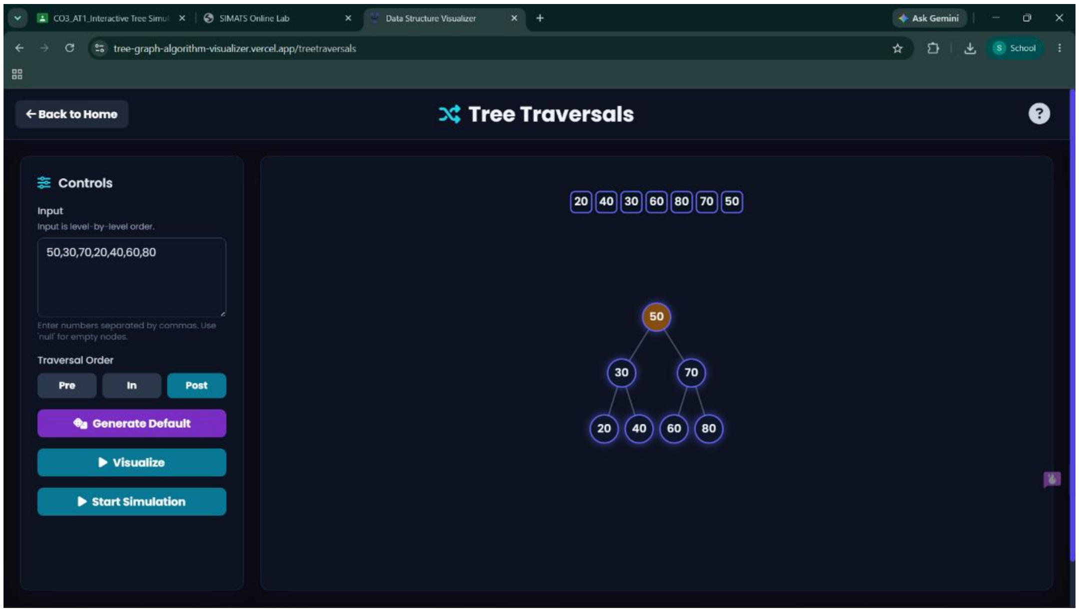
a. Inorder



b. Preorder

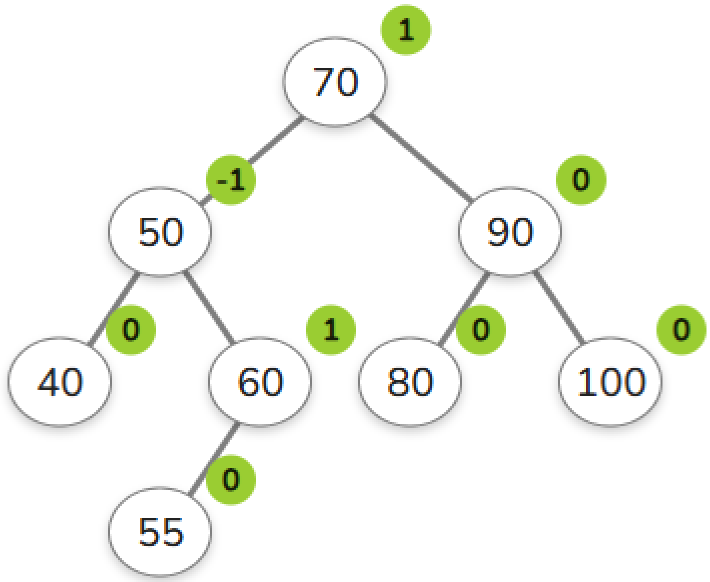


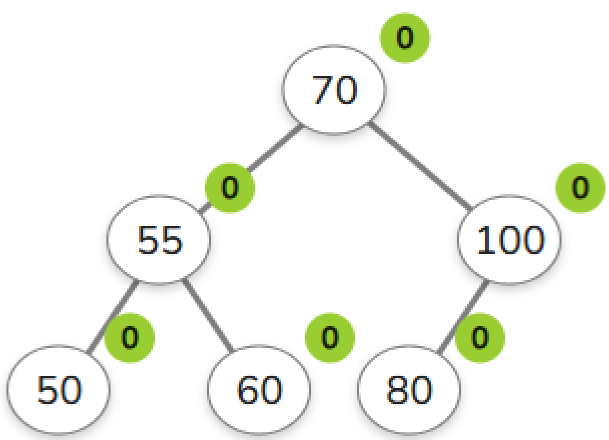
c. Postorder



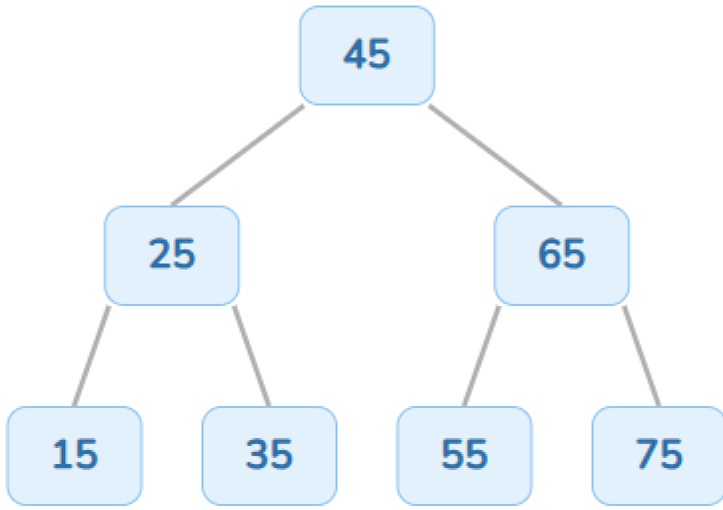
CO3_AT1_Interactive Tree Simul
SIMATS Online Lab
Data Structure Visualizer
Ask Gemini
tree-graph-algorithm-visualizer.vercel.app/treetraversals
School
Back to Home
Tree Traversals
Controls
Input
Input is level-by-level order.
50,30,70,20,40,60,80
Enter numbers separated by commas. Use 'null' for empty nodes.
Traversal Order
Pre
In
Post
Generate Default
Visualize
Start Simulation
20 40 30 60 80 70 50
50
30
70
20
40
60
80

3.Insert70,50,90,40,60,80,100,55intoanAVLTree.Delete40and90.Analyzehowthetree rebalances after each deletion.

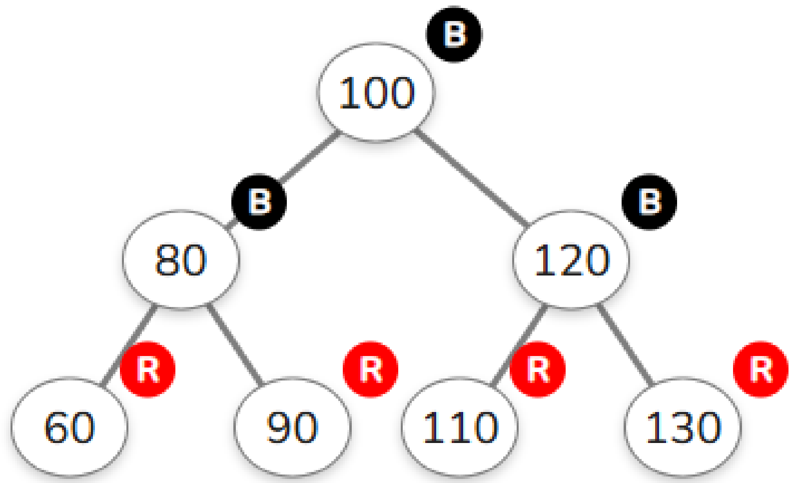




4.ConstructaB-Treeoforder3byinserting45,25,65,15,35,55,75



5. Insert100,80,120,60,90,110,130intoaRed-BlackTree.Analyzetheirheights.



Insertion	Tree Height
100	0
80	1
120	1
60	2
90	2
110	2
130	2